

WEEK – 3 ASSIGNMENT

Task Name: BA

Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)

1. It supports strategic supply chain network design.
2. It determines the most economical warehouse configuration.
3. It considers fixed and variable operating costs.
4. It ensures that production capacities are not exceeded.
5. It ensures that warehouse capacities are respected.
6. It guarantees the required customer service level.
7. It provides an optimal transportation plan.
8. It can be extended to include multiple products, transportation modes, inventory, and additional constraints.

9. Constraints

9.1 Factory Production Capacity Constraint

A factory cannot ship more products than its production capacity.

Therefore:

$$\sum_{j \in W} x_{ij} \leq S_i \quad \forall i \in F \quad \boxed{\sum_{j \in W} x_{ij} \leq S_i \quad \forall i \in F}$$

For example, for Factory 1:

$$x_{11} + x_{12} + x_{13} \leq 1000 \quad \boxed{x_{11} + x_{12} + x_{13} \leq 1000}$$

For Factory 2:

$$x_{21} + x_{22} + x_{23} \leq 1200 \quad \boxed{x_{21} + x_{22} + x_{23} \leq 1200}$$

For Factory 3:

$$x_{31} + x_{32} + x_{33} \leq 800 \quad \boxed{x_{31} + x_{32} + x_{33} \leq 800}$$

10. Warehouse Flow Balance Constraint

The amount entering a warehouse should equal the amount leaving the warehouse, assuming no inventory is carried forward.

Therefore:

$$\sum_{i \in F} x_{ij} = \sum_{k \in C} y_{jk} \quad \forall j \in W \quad \boxed{\sum_{i \in F} x_{ij} = \sum_{k \in C} y_{jk} \quad \forall j \in W}$$

For Warehouse 1:

$$x_{11} + x_{21} + x_{31} = y_{11} + y_{12} + y_{13} + y_{14} \quad \boxed{x_{11} + x_{21} + x_{31} = y_{11} + y_{12} + y_{13} + y_{14}}$$

For Warehouse 2:

$$x_{12} + x_{22} + x_{32} = y_{21} + y_{22} + y_{23} + y_{24} \quad \boxed{x_{12} + x_{22} + x_{32} = y_{21} + y_{22} + y_{23} + y_{24}}$$

For Warehouse 3:

$$x_{13}+x_{23}+x_{33}=y_{31}+y_{32}+y_{33}+y_{34} \quad x_{\{13\}}+x_{\{23\}}+x_{\{33\}} = y_{\{31\}}+y_{\{32\}}+y_{\{33\}}+y_{\{34\}}$$

This ensures conservation of material throughout the network.

11. Warehouse Capacity Constraint

Each warehouse has a maximum storage or handling capacity.

The warehouse capacity constraint is:

$$\sum_{k \in C} y_{jk} \leq U_j \quad \forall j \in W \quad \boxed{\sum_{k \in C} y_{jk} \leq U_j \quad \forall j \in W}$$

For Warehouse 1:

$$y_{11}+y_{12}+y_{13}+y_{14} \leq 1000 \quad y_{\{11\}}+y_{\{12\}}+y_{\{13\}}+y_{\{14\}} \leq 1000 \quad z_1$$

For Warehouse 2:

$$y_{21}+y_{22}+y_{23}+y_{24} \leq 900 \quad y_{\{21\}}+y_{\{22\}}+y_{\{23\}}+y_{\{24\}} \leq 900 \quad z_2$$

For Warehouse 3:

$$y_{31}+y_{32}+y_{33}+y_{34} \leq 800 \quad y_{\{31\}}+y_{\{32\}}+y_{\{33\}}+y_{\{34\}} \leq 800 \quad z_3$$

If $z_j=0$, the right-hand side becomes zero, meaning that warehouse cannot process any shipment.

12. Customer Demand Constraint

Each customer zone must receive the required amount of product.

Therefore:

$$\sum_{j \in W} y_{jk} \geq D_k \quad \forall k \in C \quad \boxed{\sum_{j \in W} y_{jk} \geq D_k \quad \forall k \in C}$$

For Customer Zone 1:

$$y_{11}+y_{21}+y_{31} \geq 600 \quad y_{\{11\}}+y_{\{21\}}+y_{\{31\}} \geq 600$$

For Customer Zone 2:

$$y_{12}+y_{22}+y_{32} \geq 700 \quad y_{\{12\}}+y_{\{22\}}+y_{\{32\}} \geq 700$$

For Customer Zone 3:

$$y_{13}+y_{23}+y_{33} \geq 500 \quad y_{\{13\}}+y_{\{23\}}+y_{\{33\}} \geq 500$$

For Customer Zone 4:

$$y_{14}+y_{24}+y_{34} \geq 400 \quad y_{\{14\}}+y_{\{24\}}+y_{\{34\}} \geq 400$$

Thus, the total demand of 2,200 units must be satisfied.

13. Minimum Service-Level Constraint

The company requires a minimum customer service level.

Let:

$$\alpha_k=0.95 \quad \alpha_k=0.95$$

for a 95% minimum service level.

The service-level constraint is:

$$\sum_{j \in W} y_{jk} \geq \alpha_k D_k \quad \forall k \in C \quad \boxed{\sum_{j \in W} y_{jk} \geq \alpha_k D_k \quad \forall k \in C}$$

For example, for Customer Zone 1:

$$y_{11}+y_{21}+y_{31} \geq 0.95(600) \quad y_{\{11\}}+y_{\{21\}}+y_{\{31\}} \geq 0.95(600)$$

$$y_{11}+y_{21}+y_{31} \geq 570 \quad y_{\{11\}}+y_{\{21\}}+y_{\{31\}} \geq 570$$

Similarly, for Customer Zone 2:

$$y_{12} + y_{22} + y_{32} \geq 0.95(700) \quad y_{12} + y_{22} + y_{32} \geq 0.95(700)$$

$$y_{12} + y_{22} + y_{32} \geq 665 \quad y_{12} + y_{22} + y_{32} \geq 665$$

The service-level constraint ensures that customers receive at least the specified percentage of their demand.

If full demand satisfaction is required, the demand constraint can be used instead:

$$\sum_{j \in W} y_{jk} \geq D_k \quad \sum_{j \in W} y_{jk} \geq D_k$$

14. Warehouse Activation Constraint

Shipments can only be sent through an active warehouse.

A Big-M formulation can be used:

$$x_{ij} \leq M_{ij} z_j \quad \forall i \in F, j \in W \quad x_{ij} \leq M_{ij} z_j \quad \forall i \in F, j \in W$$

Where M_{ij} is a sufficiently large upper bound on the shipment.

If:

$$z_j = 0$$

then:

$$x_{ij} = 0$$

Therefore, no product can be sent to a closed warehouse.

15. Non-Negativity Constraints

Transportation quantities cannot be negative.

Therefore:

$$x_{ij} \geq 0 \quad \forall i, j \quad x_{ij} \geq 0 \quad \forall i, j$$

and

$$y_{jk} \geq 0 \quad \forall j, k \quad y_{jk} \geq 0 \quad \forall j, k$$

16. Binary Constraints

Warehouse opening decisions are binary:

$$z_j \in \{0, 1\} \quad \forall j \in W \quad z_j \in \{0, 1\} \quad \forall j \in W$$

A value of 1 means the warehouse is open, while 0 means the warehouse is closed.

17. Complete MILP Formulation

The complete mathematical model is:

$$\min Z = \sum_{i \in F} \sum_{j \in W} c_{ij} F x_{ij} + \sum_{j \in W} \sum_{k \in C} c_{jk} W y_{jk} + \sum_{j \in W} f_j z_j \quad \min Z = \sum_{i \in F} \sum_{j \in W} c_{ij} F x_{ij} + \sum_{j \in W} \sum_{k \in C} c_{jk} W y_{jk} + \sum_{j \in W} f_j z_j$$

Subject to:

Factory Capacity

$$\sum_{j \in W} x_{ij} \leq S_i \quad \forall i \in F \quad \sum_{j \in W} x_{ij} \leq S_i \quad \forall i \in F$$

Warehouse Flow Balance

$$\sum_{i \in F} x_{ij} = \sum_{k \in C} y_{jk} \quad \forall j \in W \quad \sum_{i \in F} x_{ij} = \sum_{k \in C} y_{jk} \quad \forall j \in W$$

Warehouse Capacity

$$\sum_{k \in C} y_{jk} \leq U_j \quad \forall j \in W \quad \sum_{k \in C} y_{jk} \leq U_j \quad \forall j \in W$$

Customer Demand

$$\sum_{j \in W} y_{jk} \geq D_k \forall k \in C \quad \sum_{j \in W} y_{jk} \geq \alpha_k D_k \forall k \in C$$

Minimum Service Level

$$\sum_{j \in W} y_{jk} \geq \alpha_k D_k \forall k \in C \quad \sum_{j \in W} y_{jk} \geq \alpha_k D_k \forall k \in C$$

Warehouse Activation

$$x_{ij} \leq M_{ij} z_j \forall i \in F, j \in W \quad x_{ij} \leq M_{ij} z_j \forall i \in F, j \in W$$

Non-Negativity

$$x_{ij} \geq 0, y_{jk} \geq 0, z_j \in \{0, 1\}$$

Binary Condition

$$z_j \in \{0, 1\}$$

18. Constraint Matrix Representation

The MILP can be represented in standard matrix form.

Let the decision vector be:

$$X = [x \ y \ z]$$

where:

- x = factory-to-warehouse shipment variables.
- y = warehouse-to-customer shipment variables.
- z = warehouse-opening binary variables.

The general MILP formulation is:

$$\min c^T X$$

Subject to:

$$AX \leq b$$

$$EX = 0$$

$$X \geq 0$$

with:

$$z_j \in \{0, 1\}$$

The matrix A contains capacity, demand, service-level, and warehouse activation constraints.

The matrix E contains warehouse flow-balance equations.

19. Constraint Matrix Structure

The constraint matrix can be represented conceptually as:

$$\begin{bmatrix} A_F & 0 & 0 & 0 & 0 \\ A_W & -A_C & 0 & 0 & 0 \\ A_D & 0 & -M & 0 & 0 \\ A_S & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \leq \begin{bmatrix} S \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

where:

- A_F represents factory capacity.
- A_W represents warehouse inbound flows.
- A_C represents warehouse outbound flows.
- A_D represents customer demand.
- A_M represents warehouse activation.

- ASA_S represents service-level requirements.
- SS represents factory capacities.
- DD represents customer demand.
- αD represents minimum service-level quantities.

20. Example Cost Data

For practical implementation, the following illustrative transportation costs can be used.

Factory-to-Warehouse Transportation Cost

Factory	W1	W2	W3
F1	4	6	8
F2	5	4	7
F3	8	6	5

Factory W1 W2 W3

F1 4 6 8

F2 5 4 7

F3 8 6 5

Warehouse-to-Customer Transportation Cost

Warehouse	C1	C2	C3	C4
W1	3	5	6	8
W2	5	3	4	6
W3	7	6	3	4

Warehouse C1 C2 C3 C4

W1 3 5 6 8

W2 5 3 4 6

W3 7 6 3 4

Fixed Warehouse Operating Cost

Warehouse	Fixed Cost
W1	5,000
W2	4,500
W3	4,000

Warehouse Fixed Cost

W1 5,000

W2 4,500

W3 4,000

These values are illustrative and can be replaced with actual company data.

21. Solution Approach

The model can be solved using optimization software such as:

- IBM ILOG CPLEX
- Gurobi
- Python with PuLP
- Python with Pyomo
- MATLAB
- Excel Solver

The solver evaluates different combinations of warehouse-opening decisions and shipment quantities.

The optimal solution provides:

- Optimal warehouse locations.
- Factory-to-warehouse shipment quantities.
- Warehouse-to-customer shipment quantities.
- Total transportation expenditure.
- Total fixed warehouse expenditure.
- Minimum total supply chain expenditure.

22. Expected Results

After solving the MILP model, the following outputs are expected:

Decision Output

Warehouse W1 Open/Closed

Warehouse W2 Open/Closed

Warehouse W3 Open/Closed

Factory-to-Warehouse Flow Optimal quantity

Warehouse-to-Customer Flow Optimal quantity

Customer Service Level At least required percentage

Total Transportation Cost Minimum

Total Fixed Cost Minimum

Total Supply Chain Cost Minimum

The optimal solution should satisfy all factory capacities, warehouse capacities, customer demand, and service-level requirements.

23. Advantages of the MILP Model

The proposed model provides several advantages:

24. Conclusion

The Multi-Echelon Supply Chain Network Design problem can be effectively formulated as a Mixed-Integer Linear Programming model. The proposed model integrates factories, warehouses, and customer zones into a single optimization framework.

The objective function minimizes total supply chain operating expenditure, including factory-to-warehouse transportation costs, warehouse-to-customer transportation costs, and fixed warehouse operating costs.

The constraints ensure that factory production capacities and warehouse capacities are not exceeded, material flow is balanced, customer demand is satisfied, and minimum service-level requirements are maintained.

The binary warehouse-opening variables make it possible to determine the most appropriate warehouse configuration. Therefore, the MILP model provides a systematic and cost-effective approach for designing an efficient

global logistics network.

Overall, the model helps supply chain managers make better decisions regarding facility selection, transportation allocation, capacity utilization, customer service, and total operating cost minimization.